



Term Evaluation (Odd) Semester Examination September 2025

Roll no.

Name of the Course: M. Tech CSE: (I Sem)

Name of the Paper: Applied Data Science

Paper Code: (MCS-131)

Time: 1.5-hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks) (CO1)

a. How does Data Science differ from traditional statistics in terms of its interdisciplinary approach and tools used?

OR

b. Describe the typical roles and responsibilities of a data scientist, including how they collaborate with other teams in an organization.

Q2.

(10 Marks) (CO1)

a. How can data science be applied in business settings, and provide examples of use cases in industries like healthcare or e-commerce?

OR

b. Why is R considered an important programming language for data science, particularly in terms of its strengths in statistical analysis and data visualization?

Q3.

(10 Marks) (CO1)

a. Explain the basic data types and variables in R, and demonstrate with examples how operators, conditional statements, loops, and functions can be used in an R script to manipulate data.

OR

b. How discrete random variable and continuous random variable are different from one another? Justify your answer with an example of each.

Q4.

(10 Marks) (CO2)

a. A fair six-sided die is rolled. Define the discrete random variable representing the outcome of the roll and list its possible values along with their probabilities.

OR

b. Explain the concept of a probability density function (PDF) for a continuous random variable and how it differs from a probability mass function (PMF) for a discrete random variable.

Q5.

(10 Marks) (CO2)

a. In a dataset containing customer purchase records, 15% of the entries for the "purchase amount" column are missing. Discuss two appropriate methods to handle these missing values and explain how each method might impact the analysis of the dataset.

OR

b. Describe the process of collecting data from multiple sources (e.g., surveys and online transactions), staging it in a centralized database, cleaning inconsistencies (such as duplicate entries), and transforming it (e.g., normalizing numerical values). How does each step contribute to ensuring reliable data for analysis?